3 A ball is thrown horizontally from the top of a building, as shown in Fig. 3.1

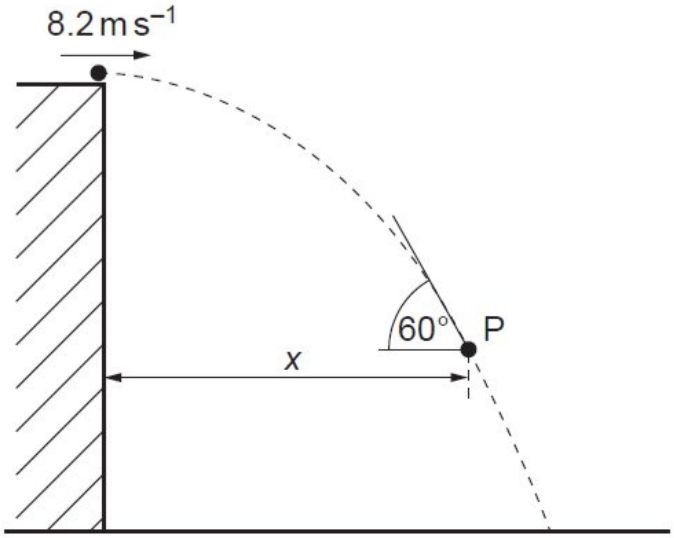


Fig. 3.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The sides of the building is vertical. At a point P on the path of the ball, the ball is at distance *x* from the building and is moving at an angle of 60° to the horizontal. Air resistance is negligible.

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

(a) Explain why the curved path is parabolic.

.....

..... [1]

(b) Calculate

(i) the time through which the ball has fallen from the top to point P

time = ... s [2]

(ii) the horizontal distance x

x = ... m [1]

(c) For the movement of the ball from the top to point P, state and explain whether the displacement of the ball is less than, more than or the same as the distance travelled by the ball.

.....

.....

..... [2]

(d) The path of the ball in **(a)**, with an initial horizontal speed of 8.2 m s^{-1}, is again shown in Fig. 3.2.

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

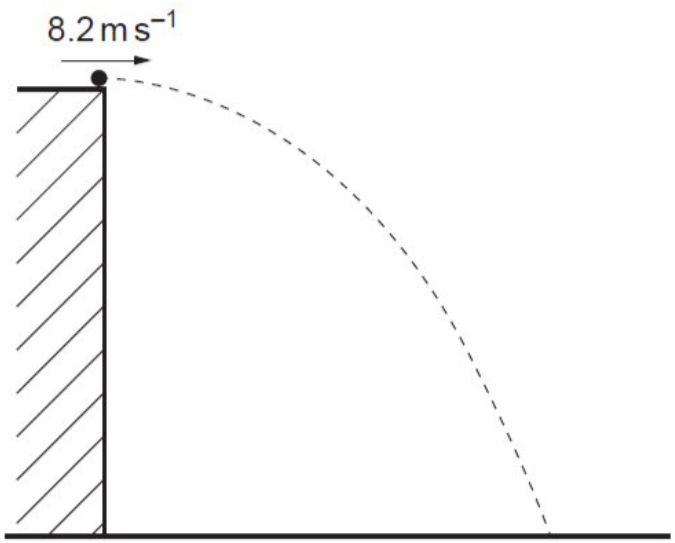


Fig. 3.2

On Fig. 3.2, sketch the new path of the ball having an initial horizontal speed

(i) greater than 8.2 m $s^{-1}$and with negligible air resistance (label this path G)

(ii) equal to 8.2 m $s^{-1}$and with air resistance (label this path A)

[3]

[Total : 9]